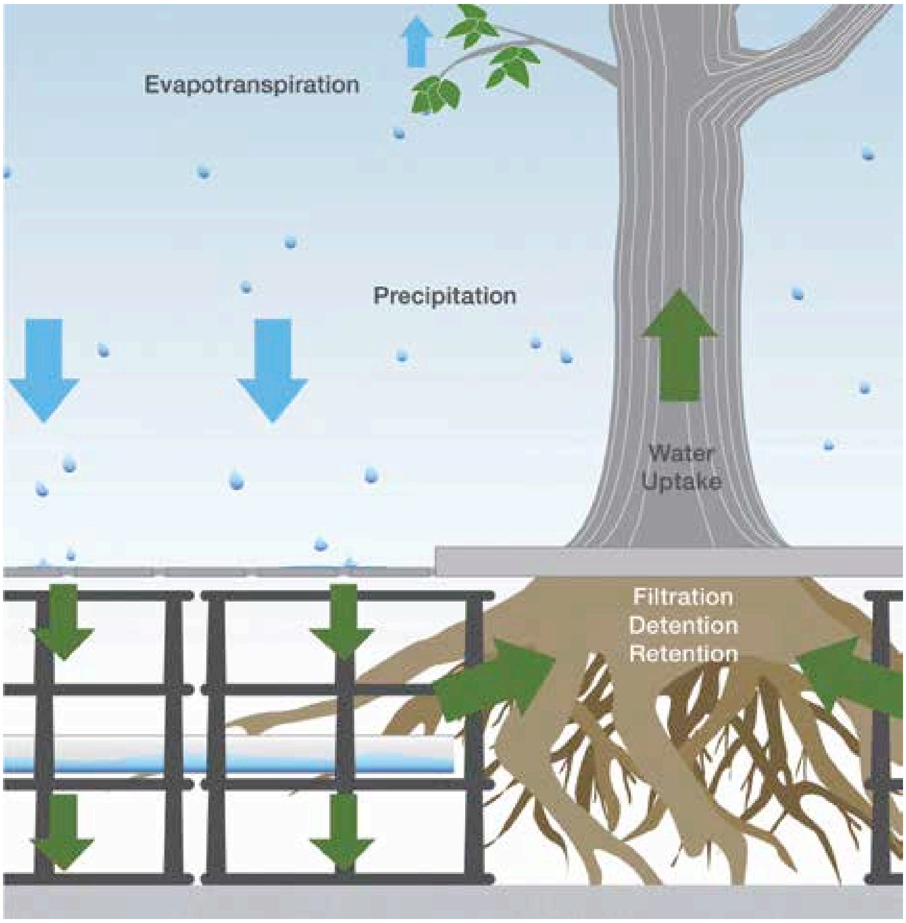
help of quinones. Still in the process of overcoming some fundamental limitations such as low energy and power density, these Harvard geniuses are close to creating something near magical in as soon as three years.

Theoretically, these flow batteries should be capable of storing one kilowatt-hour of energy using chemicals costing one-third of the price of existing systems – $27, or ₹1,652, to be precise.

Thermal-Depolymerisation

Oil – the liquid gold, over which wars have been fought and because of which countries seek separation. Up until now, it has been a purely geographical resource. Now imagine being able to mass produce oil from waste in any location. A process called 'thermal-depolymerisation' is able to convert any carbon-based waste – from turkey guts to used tires – into oil. According to sources, a ton of turkey waste can be used to produce about 600 pounds of petroleum. The waste is subjected to high pressure and heat, which decomposes long-chained polymers of hydrogen, oxygen and carbon into short-chained petroleum hydrocarbons. The process is very similar to how oil is naturally produced, except that it is fast-tracked by about a million years. It mimics a natural geological process, responsible for creating fossil fuels.



'Tis the circle of business life, explained in a way that even 7-year-olds can understand

'Water ATMs' by Sarvajal

We live in a country where millions of people still drink contaminated water, simply because it isn't feasible to build water purifying infrastructure in villages. 'Sarvajal', a charitable initiative by the Piramal Foundation, has deployed solar-powered 'water ATMs', which dispense clean water simply with the swipe of a prepaid card. The water ATMs are supplied with water by filtration systems built in larger villages. The machinery, which is scattered all over India, is maintained remotely using cloud computing and mobile technology. The company keeps tabs on water supply and demand, thus keeping the system as transparent as possible.

Learning to use the Water ATMs

TerraPower's Travelling Wave Reactor

TerraPower's Travelling Wave Reactor (TWR) is a highly advanced nuclear reactor that offers a safe and economic form of low-carbon energy. TWRs can theoretically run self sustained for decades without refueling or removal of the spent fuel. Often called 'breed and burn' reactors, they run on depleted uranium fuel.

Travelling Wave Reactors are not only a greener energy path, but also a safer one. Typically, nuclear reactors are fuelled by uranium enriched by its natural form. This process of enrichment is almost identical to the process of creating nuclear weapons. So, having an enrichment plant opens up the possibility of having nuclear weapon capabilities. Nuclear plants also need a reprocessing facility to function. This is also a potential path for nuclear weaponry. TWRs require minimal enrichment and no reprocessing and are hence completely safe from a political standpoint.

Silva Cells

'Silva Cells' are a new technology that will help us bring the forests into the cities ensuring that our concrete jungles have enough greenery.